

Lyrebird

Mathematical DSP specification and HLS hand-off

KV260 agentic-HLS demonstration

Revision 0.1 — implementation starting point

Objective. Specify a sixteen-voice synthesiser combining sine-carrier phase modulation, wavefolding, independently controlled resonant filters, envelopes, low-frequency modulation and damped reverberation. The intended range includes pure tones, harmonic and inharmonic timbres, and dynamic vocal-formant textures. Generate signed 24-bit PCM for the existing I2S path. Design the four-control mapping separately from the full internal DSP parameter space.

Specification status. The architecture below records the selected signal-processing stages. Concrete coefficients, bounds, state policies and audition patches labelled *v0 defaults* are engineering choices supplied by this hand-off to make implementation possible today; they have not been auditioned or measured. They may be revised with documented evidence. This document specifies the next DSP; it does not claim that the existing playable saw MVP already implements it. Do not silently simplify away independent envelopes, modulation routes, or the clean-sine setting to make a build pass.

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1 Signal graph and implementation boundary

For each of sixteen voices:

$$\begin{aligned} \text{PM sine} &\longrightarrow \text{wavfolder} \longrightarrow (\text{direct} + \text{BPF}_1 + \text{BPF}_2) \\ &\longrightarrow \text{amplitude ADSR} \times \text{velocity}. \end{aligned}$$

Each BPF centre has its *own* ADSR. Two per-voice LFOs have independent signed routing depths into the source, folder, filters and gain coordinates.

$$\begin{aligned} \text{Voice sum} &\longrightarrow \text{shared damped four-line FDN plus dry path} \\ &\longrightarrow \text{master gain, finite-value handling, saturation and rounding} \longrightarrow \text{existing I2S}. \end{aligned}$$

- Nominal output rate $f_s = 48000$ Hz; sixteen voices; A4 is 440 Hz in 12-TET. The accepted approximately +19.3 ppm hardware-clock error is not a redesign task.
- Target `xck26-sfvc784-2LV-c`, nominal 100 MHz HLS clock. One completed output sample per request, within **2000 clock cycles**, including all voices and reverb. No resource or timing fit is established by these equations.
- Keep the external HLS top `synth_hls`: sixteen decoded active/note/ velocity entries, four signed 24-bit Q2.22 controls, one signed 24-bit PCM output. Do not put MIDI, AXI or I2S handling in the mathematical kernel.
- Existing register software presents four Q10.22 words; RTL clamps them to Q2.22. Controls are separate from the full internal parameter object.
- Preserve `david:synth:synth_hls:1.0` and the public customisation module `synth_hls_ip`, or explicitly coordinate a versioned interface change. The generated raw module is package-private.
- Output remains mono duplicated into both stereo payloads. Real stereo is a separate interface change. Preserve scheduler, CDC, serializer and diagnostic tone.

2 Types, time and state

Let $n \in \mathbb{N}$ index output samples. Let $v \in \{0, \dots, 15\}$ index voices. Signals are sequences, e.g. $x : \mathbb{N} \rightarrow \mathbb{R}$; a memoryless nonlinearity acts pointwise, whereas a filter is a state transition with an output map.

The scalar clipping operator is

$$\text{clip}(x; a, b) = \min(b, \max(a, x)), \quad a \leq b.$$

Use finite parameters; reject nonfinite patch values before use. Define $\text{frac}(x) = x - \lfloor x \rfloor$, including for negative x .

Each voice owns: captured note/velocity; gate history; carrier and modulator phases; two LFO phases; three ADSR states; two pairs of filter integrator states; parameter smoothing state; and any explicitly selected retrigger de-click state. The reverb owns four delay buffers, pointers and four damping states. All begin at zero on a fresh PL configuration, except specified parameter defaults.

Logical sample order. Consume a coherent input snapshot and handle gate edges; evaluate current envelopes/LFOs; derive bounded physical parameters; evaluate oscillator, folder, filters, amplitude and mix; read and update reverb; convert PCM; then advance oscillator/LFO phases and envelope segments. Filter and reverb recurrences below explicitly describe their within-sample updates. Never read a partially updated voice or control snapshot.

2.1 Voice lifecycle and existing allocator

On onset capture note and velocity. Begin phases and ADSRs at zero for a fresh voice. Note-off begins all three releases from their current values; continue the captured oscillator while the amplitude release is nonzero. On reaching zero, clear the per-voice filter states and idle the voice; do not clear shared reverb.

Integration conflict to handle explicitly. Existing Linux allocation makes a released slot reusable on the next 3 ms iteration. Long releases therefore cannot reserve that slot without a control-policy change. V0 keeps this allocation policy: an unreused slot completes its release, but reuse replaces its old note. Use a short de-click transition when replacing a nonzero contribution. A concrete bounded default is to capture the last emitted voice sample, fade that held value to zero over 96 output samples, and simultaneously start the new voice with a minimum 96-sample amplitude attack. This is not preservation of the old note's full release tail. A second complete old-voice state is an optional later quality choice, not implicit extra polyphony.

Detect onset from gate edges; active note/velocity changes are also replacement events. Same-note, same-velocity retriggers need an observable gate-low snapshot. Verify the existing publish/reuse path provides this; if not, specify a control handshake or event generation extension before claiming reliable retriggering. Do not infer events which are absent from the interface. Full preservation of sixteen held voices plus arbitrary tails is not part of this sixteen-slot design.

3 Independent ADSRs

For each voice let $e_a, e_1, e_2 : \mathbb{N} \rightarrow [0, 1]$ denote amplitude and the two spectral envelopes. Every envelope has independent (T_A, T_D, S, T_R) and its own stage, elapsed counter and captured segment start. V0 uses linear segments; exponential curves can be added as explicit parameters later.

At rate F samples/s, a segment duration T has $N_T = \max(1, \text{round}(FT))$ transitions. For captured endpoints u, w :

$$e[k] = u + (w - u) \frac{k}{N_T}, \quad k = 0, \dots, N_T.$$

Emit the current value then advance. Attack goes $0 \rightarrow 1$, decay $1 \rightarrow S$; sustain holds S while gated; release goes from the current value to zero. At a segment boundary retain its endpoint as the start of the next segment; the following advance progresses along that segment. Note-off takes priority over attack/decay progression and captures the currently emitted value. Sustain zero may silence a held voice, but gate tracking must continue.

V0 bounds: $T_A, T_D, T_R \in [0.002, 8]$ seconds for amplitude; $[0.001, 8]$ seconds for spectral envelopes; $S \in [0, 1]$. Capture durations on segment entry and sustain on note-on so patch edits do not discontinuously retune a segment already in progress. Later envelope editing semantics must be explicit rather than an accidental counter discontinuity.

4 LFOs and parameter modulation

For $r = 1, 2$ and each voice, let $\ell_r[n] = \sin(2\pi\psi_r[n])$ with

$$\psi_r[n+1] = \text{frac}\left(\psi_r[n] + \frac{v_r}{f_s}\right), \quad v_r \in [0.01, 30] \text{ Hz}.$$

V0 resets both phases to zero on onset. Keep rate and initial phase independently parameterised; free-running or voice-offset phase policies are later selectable behaviours. LFOs do not modulate their own rates in v0; this keeps evaluation acyclic. Amplitude tremolo is an ordinary destination.

Let $q \in \mathbb{R}^{15}$ be dimensionless destination coordinates and $D \in \mathbb{R}^{15 \times 2}$ a fully populated signed depth matrix. For every voice:

$$q_v[n] = \bar{q}[n] + E e_v[n] + D \ell_v[n].$$

Only BPF-centre rows of E are nonzero in v0: $E_{6,2} = d_1$ and $E_{8,3} = d_2$ with one-based indices in the table below; $e_v = (e_a, e_1, e_2)^T$. The amplitude envelope separately multiplies the audio. Each of the thirty entries of D is independent; zero disables a route. d_1, d_2 are signed octaves.

Smooth user edits to base coordinates, D , and spectral depths before evaluating the expression, but do not low-pass the evaluated envelopes/LFOs a second time:

$$\bar{z}[n+1] = \bar{z}[n] + \beta(z_{\text{target}}[n] - \bar{z}[n]), \quad \beta = 1 - e^{-1/(0.005f_s)}.$$

This 5 ms edit smoother is a v0 default. Initialise it to the loaded patch, not zero. Reverb edits use 20 ms smoothing. Coefficient-rate optimisations must retain these audible trajectories within the numerical acceptance budget.

Coordinate	Physical decoding	V0 range after decoding
1	carrier $f_c = f_N 2^{q_1}$	[1, 12000] Hz
2	PM ratio $\rho = 2^{q_2}$, $f_m = f_c \rho$	$\rho \in [1/16, 16]$, $f_m \in [0, 18000]$ Hz
3	PM depth $I = q_3$	[0, 12] radians
4	fold drive $g = 2^{q_4}$	[1, 16]
5	fold wet $w = q_5$	[0, 1]
6	BPF centre $f_1 = 440 \cdot 2^{q_6}$	[40, 12000] Hz
7	BPF $Q_1 = 2^{q_7}$	[0.5, 12]
8	BPF centre $f_2 = 440 \cdot 2^{q_8}$	[40, 12000] Hz
9	BPF $Q_2 = 2^{q_9}$	[0.5, 12]
10–12	raw branch weights $a_0 = q_{10}, a_1 = q_{11}, a_2 = q_{12}$	each $[-2, 2]$
13	voice trim $t = 2^{q_{13}}$	[0, 2]
14	amplitude modulation multiplier $h = q_{14}$	[0, 1]
15	extra PM frequency offset $\Delta f_m = 1000 q_{15}$	$[-8000, 8000]$ Hz

For row 2 include row 15 before clamping: $f_m = \text{clip}(f_c \rho + \Delta f_m; 0, 18000)$. Clamp exponent coordinates before exponentiation, using the inverse of the stated bounds, to avoid overflow. Voice trim uses $q_{13} \in [-16, 1]$ (zero is obtained through the envelope or h , not $\log 0$). The table's row-2 ratio/frequency and row-15 offset are independent source controls, not four-dial assignments. V0 envelope depths lie in $[-6, 6]$ octaves; matrix entries are finite and in $[-8, 8]$ coordinate units. Destination clipping defines valid parameters, not a proof of stability under arbitrary time variation; stress-test the whole range.

5 Phase-modulated sine and wavefolder

For captured MIDI note N , $f_N = 440 2^{(N-69)/12}$. In cycles, phases $\theta_c, \theta_m \in \mathbb{R}/\mathbb{Z}$ obey

$$\begin{aligned} x_v[n] &= \sin(2\pi\theta_c[n] + I[n] \sin(2\pi\theta_m[n])), \\ \theta_c[n+1] &= \text{frac}(\theta_c[n] + f_c[n]/f_s), \\ \theta_m[n+1] &= \text{frac}(\theta_m[n] + f_m[n]/f_s). \end{aligned}$$

Equivalently, for $c, m \in S^1 \subset \mathbb{C}$, the readout is $\Im(c \exp(iI\Im m))$; each state rotates by its corresponding frequency. This is sine-carrier PM, not a saw readout. $I = 0$ gives a pure sine. Noninteger modulator/carrier ratios permit inharmonic sidebands.

Define the triangular fold $\mathcal{F} : \mathbb{R} \rightarrow [-1, 1]$ by

$$r(u) = u + 1 - 4 \left\lfloor \frac{u+1}{4} \right\rfloor, \quad \mathcal{F}(u) = 1 - |r(u) - 2|.$$

Then $\mathcal{F}(u) = u$ for $u \in [-1, 1]$ and the period is 4. The folded signal is

$$z_v[n] = (1 - w[n])x_v[n] + w[n]\mathcal{F}(g[n]x_v[n]).$$

For example $\mathcal{F}(1.2) = 0.8$, $\mathcal{F}(2) = 0$, $\mathcal{F}(3) = -1$. $g = 1$ or $w = 0$ bypasses folding exactly on the oscillator range. Do not implement negative remainder using a language operator with different floor semantics.

6 Two parallel resonant band-pass filters

V0 realisation: topology-preserving state-variable filter (trapezoidal integrator form), independently instantiated for $j = 1, 2$. Each has real states s_1, s_2 and consumes $z_v[n]$. At processing rate F define

$$g_f = \tan(\pi f_j / F), \quad k = 1/Q_j, \quad a_1 = \frac{1}{1 + g_f(g_f + k)}, \quad a_2 = g_f a_1, \quad a_3 = g_f a_2.$$

Using *old* states on every right-hand side:

$$\begin{aligned} v_3 &= z_v[n] - s_2, & v_1 &= a_1 s_1 + a_2 v_3, \\ v_2 &= s_2 + a_2 s_1 + a_3 v_3, & b_j[n] &= k v_1, \\ s_1^+ &= 2v_1 - s_1, & s_2^+ &= 2v_2 - s_2. \end{aligned}$$

The factor k gives unity centre-frequency gain in the fixed-coefficient ideal filter, so increasing Q narrows the peak rather than automatically multiplying its centre gain. Time-varying transients still require headroom and testing. Do not algebraically reuse updated integrator state in the same sample.

Normalise signed branch weights to bound their combined static gain:

$$\hat{a}_j = \frac{a_j}{\max(1, |a_0| + |a_1| + |a_2|)}, \quad r_v[n] = \hat{a}_0 z_v[n] + \hat{a}_1 b_1[n] + \hat{a}_2 b_2[n].$$

Do not cascade the BPFs. Their centres and ADSRs must remain independent. Filter state must still evolve while its branch weight is zero if it may be faded back in; bypass alone is not permission to retain stale state indefinitely.

7 Amplitude and mixing

With captured velocity $V_v \in \{1, \dots, 127\}$:

$$o_v[n] = \frac{V_v}{127} e_{a,v}[n] t_v[n] h_v[n] r_v[n], \quad x[n] = \frac{1}{16} \sum_{v=0}^{15} o_v[n].$$

The $1/16$ is fixed, not divided by the number of active notes. Add any explicitly specified replacement de-click contribution before the sum. A balanced or grouped reduction is allowed with tolerance-bounded floating-point reassociation. Velocity-zero note-on remains note-off in Linux. Do not invent voice stealing.

8 Shared four-line damped feedback delay network

Let $u, d, r : \mathbb{N} \rightarrow \mathbb{R}^4$. Delay lengths in output samples are

$$(L_1, L_2, L_3, L_4) = (1493, 1601, 1867, 2053).$$

These fixed lengths are audition starting values, not a claim of ideal diffusion. At 48 kHz the four FDN lines total 7014 single-precision delay entries, 28056 data bytes before memory packing overhead. Keep the FDN at the output rate if the voice path is oversampled. No clearing of the FDN on ordinary note events.

The input is diffused before injection by two serial allpasses of lengths 113 and 337 samples with coefficients $a_1 = 0.65$ and $a_2 = 0.55$, the second driven by the first's output $p[n]$. Each stage reads the delayed value v and, with zero initial line contents, uses

$$y = v - ax, \quad \text{write} = x + ay,$$

where write is the value stored back into the stage's delay line.

Use the fixed orthogonal matrix and unit-norm injection/readout vectors

$$A = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}, \quad b = \frac{1}{2}(1, 1, 1, 1)^T, \quad c = \frac{1}{2}(1, -1, 1, -1)^T.$$

Read all delayed values before writing any current inputs:

$$\begin{aligned} d_j[n] &= u_j[n - L_j], & u_j[n] &= 0 \text{ for } n < 0, \\ r_j[n] &= (1 - \alpha)d_j[n] + \alpha r_j[n - 1], \\ u[n] &= D b p[n] + A \text{diag}(g_1, g_2, g_3, g_4) r[n], \\ w_R[n] &= c^T d[n]. \end{aligned}$$

Set $r[-1] = 0$. Circular buffers with one read and one write per line per output sample implement these delays. The *in-loop* dampers shorten bright tails; an output LPF alone does not change the network's internal decay modes.

The single control $D \in [0, 1]$ sets the reverb T_{60} from logarithmic anchors of 0.35, 1.4, 6, 30 and 240 s at 0, 0.25, 0.5, 0.8 and 1, with $\log T_{60}$ linear in D between consecutive anchors. Per line

$$g_j = 10^{-3L_j/(f_s T_{60})}, \quad \alpha = \exp(-2\pi f_d / f_s), \quad f_d = 6000 + 12000D \text{ Hz},$$

with the same damping cutoff on all four lines. The cutoff formula is a matched-pole parameterisation, not an exact digital -3 dB specification. T_{60} is the nominal loss-only decay parameter; damping makes actual decay frequency-dependent. For fixed coefficients, orthogonal mixing, unit-gain passive dampers and $g_j < 1$ give a lossy loop. This does not excuse numerical overflow or guarantee the behaviour of later nonlinear feedback edits.

HLS implements the per-line gains from a 17-position coefficient table in D with interpolation between adjacent positions; the sampled maximum absolute gain error against the continuous mapping is 0.00240931111.

Injection is $\frac{1}{2}D$ times the diffused input $p[n]$; the wet return gain is $D(1 + 3D)$, the dry path is additive:

$$y[n] = x[n] + D(1 + 3D) w_R[n].$$

At $D = 0$ the injection and the return are both zero, a pure dry passthrough. The output then receives the fixed output tail: $8\times$ makeup, a 4:1 compressor above 0.5, then a fixed nonresonant 7 kHz two-pole LPF before PCM.

9 PCM, numerics and aliasing

Map NaN output to zero, clamp finite output and infinities to $[-1, 1]$, multiply by 2^{23} , round nearest-even, and saturate to $[-2^{23}, 2^{23} - 1]$. Do not permit a NaN to persist in feedback state: a nonfinite internal state must fail a host test, and the hardware implementation must recover or flag it by an explicitly tested policy. Ordinary saturation is not a substitute for headroom.

Use a double-precision host reference. Initial HLS may use float, LUTs or fixed point with explicit ranges and approximation error. Share coefficient arithmetic and schedule operations before deleting synthesis features. HLS must synthesise bounded storage and loops; no heap allocation or general matrix-exponential runtime is required by this DSP.

Aliasing is an explicit experiment, not a hidden promise. PM and folding generate frequencies beyond Nyquist. V0 mathematical reference runs at 48 kHz and is intentionally not claimed bandlimited. Also render the oscillator/folder/ filter voice path at $4f_s$ or $8f_s$, evaluate envelopes and LFOs on that finer time grid, and low-pass/decimate with a documented filter as an audition reference. Compare bright high notes and extreme drive. Select hardware oversampling and anti-alias filtering from listening and measured cost; include it in the 2000-cycle budget. If the first hardware demo stays at 1x, label that limitation in its report; do not describe it as transparent or alias-free. Oversampling is not automatic permission to change envelope times, phase increments or the reverb clock.

10 Reference audition fixtures

All seeds below are *unheard starting points*. Supply host-side full-parameter patch loading independently of the four external controls. Common defaults: velocity 100, voice trim 1, amplitude multiplier 1, LFO depths zero, spectral depths zero, $\Delta f_m = 0$, master gain 0.8. Unused oscillator ratio defaults to 1; unused BPFs still have valid centres 700/1700 Hz and $Q = 3$. Unless overridden, use amplitude ADSR (0.005, 0.1, 1, 0.1), both spectral ADSRs (0.060, 0.22, 0.10, 0.15), and LFO rates 3.1/4.7 Hz with zero initial phases. Spectral envelopes remain harmless when their excursion depths are zero.

Pure sine. $I = 0$, $g = 1$, $w = 0$, $(a_0, a_1, a_2) = (1, 0, 0)$, $\mu = 0$. Amplitude ADSR (0.005, 0.1, 1, 0.1) in seconds/level. Check the steady-state sine after attack, not the spectrum of its onset window.

Folded/metallic. $I = 3$, $\rho = \sqrt{2}$, $g = 4$, $w = 0.7$, $(a_0, a_1, a_2) = (0.5, 0.3, 0.2)$; retain the room seed. This is a texture target, not the claim that every such combination is musically useful.

Dry vocal-formant. Start at MIDI 60 and also audition 48 and 72. $I = 1.5$, $\rho = 1$, $g = 1.8$, $w = 0.4$; branch weights (0.15, 0.5, 0.35); $Q_1 = 4$, $Q_2 = 5$; $\mu = 0$. Amplitude ADSR (0.008, 0.25, 0.65, 0.18). First spectral ADSR (0.060, 0.22, 0.10, 0.15) with $f_{1,\text{base}} = 450$ Hz and $d_1 = \log_2(1200/450)$. Second spectral ADSR (0.035, 0.30, 0.15, 0.20) with $f_{2,\text{base}} = 1800$ Hz and $d_2 = \log_2(900/1800) = -1$. Use a 350 ms gate, then let the release finish. Formants are not forced to follow note pitch. Keep LFO depths zero first. Assess articulation through listening tests in addition to spectral analysis. A pitch scoop may be added later as a separate route/envelope; it is not silently wired into this three-envelope baseline.

Modulated vocal-formant. Starting from the dry vocal-formant fixture, use LFOs 3.1 and 4.7 Hz and small independent depths into the two centre-frequency coordinates. Begin at 0.1 octave per route, then audition increasing signed depths. Preserve the dry fixture.

Render short labelled WAVs and a compact parameter dump for every audition. Include one held note and one chord; loudness-match comparisons rather than mistaking extra gain for improved timbre.

11 Four dials and two strips: a separate mapping layer

The internal parameter object includes q , all three ADSRs, both LFO settings, D , spectral depths, reverb controls and master level. Let its dimension be d ; do not pretend $d = 4$ or confuse it with the 15 modulation destinations. The kernel accepts the fully decoded object in host/reference tests. In fabric a pure mapping function can derive that object from the four existing controls and compiled patch data without transporting every coefficient across AXI.

Fixed four-control contract. The four controls are dimensionless deviation values: A maps body, B maps edge/material, C maps pitch-stable gesture and D maps space. This is a fixed mapping, not a compiled audition patch. All-zero controls mean zero PM, fold bypass, direct branch unity, zero filter branches, all LFO routes zero and zero reverb injection and return.

Control A adds $1.2A$ radians of PM and sets the filtered mix to $0.85A$ with a $0.55/0.45$ split between the two filter branches; its formants move from 350 to 1300 Hz and from 2800 to 900 Hz, and Q moves logarithmically from 1.2 to 6. Control B adds $7B^2$ radians of PM, sets fold drive to 2^{3B^2} and fold wet to $0.85B$. Control C has no carrier-pitch, PM-ratio or absolute-Hz modulator-offset route; it produces bounded, destination-specific motion of PM, fold, formants, branches and amplitude only. Control D sets the reverb T_{60} from logarithmic anchors of 0.35, 1.4, 6, 30 and 240 s at 0, 0.25, 0.5, 0.8 and 1; at zero it gives zero injection and zero return, the dry path is additive, and the dramatic endpoint is preserved.

The fixed output tail is $8\times$ makeup, a 4:1 compressor above 0.5, then a fixed nonresonant 7 kHz two-pole LPF before PCM. The touch strips require a future versioned six-control transport after measured MIDI decoding: the left strip is channel-0 pitch bend returning to zero and the right strip is channel-0 CC1 returning to zero, with endpoints and interleaving still incomplete. Keep full parameter access in host tests.

11.1 The proposed outward rotation

For dial vector $x \in \mathbb{R}^4$, let $M_0 : \mathbb{R}^4 \rightarrow \mathbb{R}^d$ be injective and $V = \text{im } M_0$. A strip may select

$$q_{\text{latent}}(s) = \exp(sA)M_0x, \quad A^T = -A.$$

This rotates the four-dimensional image, retaining $M(s)^T M(s) = M_0^T M_0$. An especially interesting choice, for $d \geq 8$, is an isometry $J : V \rightarrow V^\perp$ and

$$q_{\text{latent}}(s) = \cos(s)M_0x + \sin(s)J(M_0x).$$

It initially moves purely out of the ordinary dial image. Define $Av = Jv$, $A(Jv) = -v$ and zero on the remaining orthogonal complement to generate this path. An affine origin and bounded nonlinear physical decoding follow as separate maps. This is a future mapping-stage design, not part of the v0 DSP recurrence or an excuse to compute a dense exponential per audio sample.

The FC25 provides two touch strips, not yet HIL characterised in this project. Capture message type, range, resolution and release behaviour before specifying their transport or normalisation. Do not assume CC1, pitch-bend format or centre return. Specific filter/pitch/mapping routes remain mapping-stage decisions. The existing four-parameter ABI does not already carry two extra independent strip signals.

12 Agent execution plan and acceptance

1. Read this source and the local knowledge-tree spec/state. Preserve the working MVP and unrelated uncommitted changes. Create a complete parameter type, state type and independent double reference with the exact ordering above.
2. Render the four fixtures. Unit-test fold identities, PM zero-depth, ADSR lengths and early release, two BPF impulse/sine responses, and FDN decay. Evaluate the vocal-formant fixture before hardware implementation. Report any departure from v0 defaults in a small parameter/decision record.
3. Implement HLS while preserving external boundaries; update tests whose old saw/ramp assertions are deliberately superseded. Keep register, scheduler, PCM, CDC and I2S coverage. Test release reuse and same-note retrigger delivery.
4. Run native tests and the established HLS sequence, inspect reports, then integrate packaged IP and run top simulation before the complete Vivado build.

5. Package and coordinate a board reload with the hardware operator. Confirm register identity, coherent controls, notes/releases, zero overruns and the reference audio behaviours. Preserve the recoverable working bitstream and service. Do not interrupt playing unannounced.

12.1 Required evidence

- Exact fold samples, periodicity, odd symmetry and bypass; pure sine at zero PM/folding with direct-only, dry output. Test negative inputs and fold boundaries.
- Every ADSR independent, correct duration to within one processing tick; release during attack/decay, repeated notes, idle, all sixteen held, release reuse.
- Filters: fixed-coefficient unity peak within numerical tolerance, centre/Q sweep, opposing envelope directions, maximum routing depths, finite states and bounded outputs over long silence and parameter-edge stress cases.
- Reverb impulse response and bandwise decay, read-before-write correctness, tail after all notes off, longest-decay finite-state stress, fresh-load silence. Do not claim exact T_{60} in every band with damping enabled.
- Suggested initial float-vs-double short-vector error target: $\text{RMS} < 10^{-4}$, $\text{peak} < 10^{-3}$ normalised full scale for two-second seeded fixtures before PCM. If long recursive tails need different error metrics, report and justify them; do not silently loosen tests. Independently test nearest-even PCM boundaries.
- Host references are not bit-accurate RTL proof. C/RTL co-simulation must match the synthesised implementation's expected PCM outputs under a documented numeric contract. Approximate transcendental functions need measured error.
- HLS worst-case latency ≤ 2000 cycles at 10 ns, sustained one request per audio sample, utilisation and memory evidence. Full implementation nonnegative timing slack and existing DRC/methodology/CDC gates; board overrun count zero.
- Separate compiler/PDF success, reference success, HLS success, routed timing, and physical audio acceptance. Successful compilation does not establish perceptual quality.

12.2 Existing implementation routes

hls/src/synth_hls.cpp	current replaceable sample implementation
hls/include/synth_hls.h	external top declaration
hls/tests/	host tests / reference owners
hls/run_hls.tcl	Vitis 2025.2 component flow
rtl/synth_hls_adapter.sv	stable RTL/HLS boundary
software/src/synth_live.c	coherent startup / 3 ms runtime
scripts/run_hls.sh	csim, synth, cosim, package
scripts/run_stage5_sim.sh	integration simulation
scripts/build_synth.sh	complete implementation gates
scripts/package_synth.sh	deployable PL_FLAT bundle

Established HLS order (csim and synth reset the component):

```
make -C hls test
scripts/run_hls.sh csim
scripts/run_hls.sh synth
scripts/run_hls.sh cosim
scripts/run_hls.sh package
```

Inspect `hls/synth_hls_proj/hls/syn/report/synth_hls_csynth.rpt` and packaged IP, not just command exits. Run the repository's existing simulation and build wrappers according to their checked usage. The accepted full-build marker is `SYNTH_BUILD_PASS`.

Deliverable: a reproducible implementation of the reference patches on the existing I2S hardware, including clean sine and dynamic vocal-formant outputs, with independent DSP parameters and documented numerical, timing and audio validation. Tune the four-dimensional control mapping after validating the DSP.